PRIVATE ASSESSMENT NOTES

Radar-Camera Fusion Project

Confidential

October 31, 2025

Document Information

Date: October 31, 2025
Subject: Candidate Performance Assessment
Purpose: Internal reference for supervisor/advisor use only

CONFIDENTIAL — For Supervisor Eyes Only

Genuine Strengths

Technical Breadth

Demonstrates competence across radar signal processing, computer vision, deep learning, and systems engineering. This is legitimately strong.

Problem Articulation

Can clearly identify and articulate technical challenges. The sensor fusion motivation is well-reasoned.

Communication

Presentations are organized and comprehensible. Monthly reports delivered consistently.

Hands-On Work

Actually built hardware rig and collected real data rather than pure simulation.

Red Flags / Genuine Concerns

[CRITICAL] Integration Incomplete

What I observed:

- CFAR detector works in isolation
- Multi-gain methodology conceptualized but not fully integrated
- CNN models "in development" but no performance results shown
- Calibration identified as critical but solutions not demonstrated
- End-to-end pipeline still "under integration" in October

Real concern:

7 months in, core system isn't assembled. For a 12-month project targeting completion, this is worrying. By month 10, the detection+labeling pipeline should be demonstrable.

Watch for:

- Promises of imminent integration that keep getting delayed
- Components working "in isolation" but failing when connected
- Excuses about external dependencies (hardware, lab access)
- Moving to new experiments instead of completing integration

Action item:

Require a clear milestone for "working end-to-end system" with specific, measurable outputs.

[CRITICAL] No Quantitative Validation

What I observed:

- Claims that multi-gain approach is "better" but no metrics
- CNN development ongoing but zero comparison to CFAR baseline
- MAE self-supervised learning shown as macro-level output ("¡10 minutes on MacBook Air") not actual detection performance
- Labeling efficiency improvements stated as theoretical benefit, not measured

Real concern:

In ML/signal processing, if you can't measure it, you haven't proven it. The entire value proposition rests on claims that aren't validated.

Red flag patterns:

- "Expected to improve detection rates" (not demonstrated)
- "Should reduce labeling time" (not measured)
- Preliminary results without error bars or statistical tests
- No ablation studies showing what actually helps

Watch for:

- Continued work without comparative benchmarks
- Excuses about "still tuning" or "not ready to publish results"
- Moving goal posts on what success looks like
- Refusing to compare against clear baselines

Action item:

Require quantitative metrics before any next phase. CNN accuracy vs. CFAR on test set. Labeling time per image with/without radar cues. Clear, comparative numbers.

[MODERATE] Scope Creep Without Clear Priorities

What I observed:

- Core system (detection + labeling) still incomplete
- Meanwhile exploring: integer representations, entropy analysis, GPU sharding, Masked Autoencoders, Vision Transformers, continual learning
- Multiple presentations on side topics that don't clearly feed core objectives
- Future directions (self-supervised learning, physics-informed models) detailed while present work unfinished

Real concern:

Looks like either:

1. Avoiding hard integration work by exploring "cooler" ML techniques, OR
2. Genuinely uncertain about priorities and spreading effort too thin, OR
3. Pursuing individual interests rather than thesis milestones

Watch for:

- Justifications like "exploring multiple approaches" when you really mean "not focused"
- Lots of presentations but little demonstrable progress on core system
- Starting new experiments before finishing old ones
- "Parallel paths" that duplicate effort

Action item:

Enforce scope. Core thesis is: (1) Detect with radar-CNN, (2) Assist image labeling. Everything else is secondary. Require rationale for why side explorations matter.

[MODERATE] Calibration Never Really Addressed

What I observed:

- Identified as "critical" for projecting radar onto camera coordinates
- Planning document mentions "autonomous calibration approach"
- No results shown on how calibration errors affect detection
- July presentations mention "calibration work" but no outcomes documented

Real concern:

This is possibly the most important technical challenge. Without good calibration, the entire fusion approach fails. Seven months in, still vague.

Watch for:

- "We'll solve calibration in Q4" promises repeated
- Workarounds to avoid the hard problem (e.g., manual calibration for testing)
- Dismissing calibration errors as "minor" without data
- Moving to synthetic experiments to avoid calibration burden

Action item:

Require clear calibration results with measured error metrics before accepting detection+labeling claims.

[MODERATE] Inconsistent Documentation Depth

What I observed:

- Some monthly reports are thorough; others are high-level summaries
- Some presentations have real data; others show conceptual diagrams only
- Failure cases not typically analyzed or discussed
- Mixed quality in what counts as "completion"

Real concern:

Hard to assess actual progress vs. narrative progress. When documentation is sparse, it's hard to distinguish real blockers from inattention.

Watch for:

- Avoiding detailed write-ups of failed experiments
- Vague language ("working on," "exploring," "planning")
- Not documenting what didn't work or why
- Selective reporting of results

Action item:

Require structured reporting: what was attempted, what worked, what failed, why, and next steps.

[MODERATE] Unclear Problem-Solving When Blocked

What I observed:

- Challenges identified (e.g., "CFAR limitations in heavy clutter," "calibration difficulty")
- Solutions sometimes proposed theoretically
- When obstacles arise, recovery path not always clear

- Some explorations (entropy, integer processing) appear to be "let's see if this helps" without clear hypothesis

Real concern:

Distinguishes between researchers who systematically solve hard problems vs. those who jump to next thing when stuck.

Watch for:

- Switching topics when hitting obstacles rather than solving them
- Lack of concrete diagnostic/debugging steps documented
- Vague statements about why things didn't work
- No evidence of systematic experimentation

Action item:

Require problem-solving methodology. When something fails, document: hypothesis, test, result, analysis, next attempt.

[MODERATE] Possible Overconfidence in Future Work

What I observed:

- Current system not yet functional, but extensive plans for:
  - Self-supervised learning with Masked Autoencoders
  - Physics-informed neural networks
  - Vision Transformers for radar
  - Continual learning strategies
  - A-MARV dataset (multi-location expansion)

Real concern:

"I'll fix the integration later; first let me explore advanced techniques" is a common pattern of unsuccessful researchers. Completing v1 is harder than planning v2.

Watch for:

- Getting excited about advanced topics while basics unfinished
- Treating planned work as if it's de facto going to happen
- Not prioritizing ruthlessly

Action item:

Require completion of core system first. Advanced features only if core system works.

Collaboration & Communication Notes

Positive:

- Onboarding undergrads (Cancilla, Kirk) shows leadership
- Regular presentations and reporting

Caution:

- Undergrads onboarded to incomplete system — may inherit unclear foundation
- Leadership demonstrated but unclear how complete projects are delegated vs. how they're mentored through challenges

Watch for:

Whether team members actually understand the work or just executing handed-off tasks.

Overall Assessment for Recommender

Dimension	Rating	Notes
Technical Knowledge	⋆⋆⋆⋆⋆	Genuine breadth
Execution	⋆⋆⋆○○	Good start, integration stalling
Rigor	⋆⋆⋆○○	Conceptual strong, empirical weak
Problem-Solving	⋆⋆⋆○○	Can identify problems, unclear on solving them
Communication	⋆⋆⋆⋆○	Clear, but documentation depth varies
Focus/Prioritization	⋆⋆⋆○○	Too many parallel explorations
Completion	⋆⋆⋆○○	50-60% of planned work, core incomplete

Key Questions to Ask Before Recommending

1. "Can you show me the end-to-end system working on your test data?"

What you want: Radar→detection→camera overlay with measured accuracy
Red flag: "Almost ready" or "still integrating"

2. "What are the actual detection rates for your CNN vs. CFAR baseline?"

What you want: Numbers (precision, recall, F1, detection probability)
Red flag: "It should be better than" or "preliminary results show"

3. "What specific technical problems did you hit and how did you solve them?"

What you want: Concrete examples of diagnosis and solution
Red flag: Vague answers or "I moved on to something else"

4. "Why are you exploring Vision Transformers when the basic CNN isn't validated yet?"

What you want: Clear answer about priority and dependencies
Red flag: "It's interesting" or "for future work" (do future work later)

5. "Show me your calibration validation. What's your co-registration error?"

What you want: Measured metrics, error analysis
Red flag: Vague promises or workarounds

6. "What would failure look like for this project? What would make you pivot?"

What you want: Clear success/failure criteria
Red flag: "I'm not worried" or all outcomes are "interesting"

Recommendation Guidance

If core system is working and validated:

Recommend with note about focus and completion.

If core system still incomplete:

Do NOT recommend. This is a research project, not a requirements collection. Being 70% done with many things is worse than 100% done with one thing.

Conditional recommendation:

"Recommend contingent on completion of end-to-end detection+labeling system with quantitative validation by [date]."

Tone for Recommendation Letter

If writing a real letter, you can be honest without being harsh:

"The candidate demonstrates strong technical breadth and identifies important research problems. Their work on sensor fusion is well-motivated. However, they are most valuable when focused on completing core objectives rather than exploring multiple advanced techniques in parallel. I recommend them for roles valuing deep technical knowledge, with the caveat that they will benefit from working with advisors who emphasize project completion and quantitative validation before scope expansion."

This is honest, not secretly negative, and actually helpful to the recipient.